

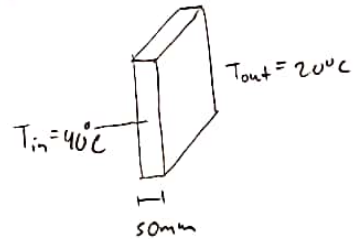
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CHE 312

Homework #2

75

1.6



$$T_{in} = 40^\circ\text{C} \quad l = 0.05\text{m}$$

$$T_{out} = 20^\circ\text{C}$$

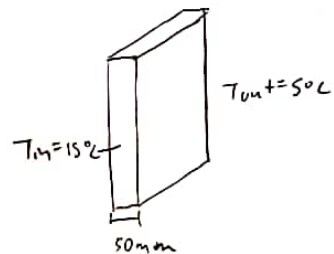
$$q'' = 40\text{ W/m}^2$$

$$q'' = k(T_{in} - T_{out})$$

$$k = \frac{q'' l}{T_{in} - T_{out}} = \frac{(40\text{ W/m}^2)(0.05\text{m})}{20^\circ - 40^\circ\text{C}}$$

$$k = -0.1\text{ W/m}\cdot\text{C}$$

1.7



$$R_{t, \text{cond}} = 1.19 \times 10^{-3}\text{ K/W}$$

$$T_{in} = 15^\circ\text{C}$$

$$T_{out} = 50^\circ\text{C}$$

$$l = 0.05\text{m}$$

$$A = 3\text{ m}^2$$

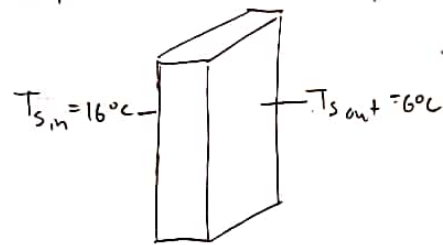
$$q = \frac{(T_{in} - T_{out})}{R_t} = \frac{(15^\circ\text{C} - 50^\circ\text{C})}{1.19 \times 10^{-3}\text{ K/W}}$$

$$q = 8403.36\text{ W}$$

$$\frac{q}{A} = \frac{k(T_{in} - T_{out})}{l} \Rightarrow k = \frac{q}{A} \cdot \frac{l}{T_{in} - T_{out}}$$

$$k = \frac{8403.36\text{ W}}{3\text{ m}^2} \cdot 0.05\text{m} \cdot \frac{1}{15^\circ\text{C} - 50^\circ\text{C}} = 14.01\text{ W/m}\cdot\text{K}$$

1.14



$$T_{\text{air, int}} = 20^\circ\text{C}$$

$$T_{\text{air, ext}} = 5^\circ\text{C}$$

$$h_{\text{in}} = 5 \text{ W/m}^2\cdot\text{K}$$

$$h_{\text{out}} = 20 \text{ W/m}^2$$

$$q'' = h(T_s - T_\infty)$$

- heat flux from interior air to the wall

$$T_{\text{air, int}} = 20^\circ\text{C}$$

$$T_{s, \text{in}} = 16^\circ\text{C}$$

$$h_{\text{in}} = 5 \text{ W/m}^2\cdot\text{K}$$

$$q'' = \frac{5 \text{ W}}{\text{m}^2\cdot\text{K}} (20^\circ\text{C} - 16^\circ\text{C}) = 20 \text{ W/m}^2$$

- heat flux from wall to exterior air

$$T_{s, \text{out}} = 6^\circ\text{C}$$

$$T_{\text{air, ext}} = 5^\circ\text{C}$$

$$h_{\text{out}} = 20 \text{ W/m}^2$$

$$q'' = 20 \text{ W/m}^2 (6^\circ\text{C} - 5^\circ\text{C}) = 20 \text{ W/m}^2$$

- heat flux from wall to interior air

$$= - (\text{heat flux from interior air to the wall.})$$

$$= -20 \text{ W/m}^2$$

The wall is at steady state b/c energy is not changing with time.